Payment Debited, Order Not Placed: Refund Options

How Paytm Documentation defines refunds and how V Foods executes them seamlessly

The Real-World Situation

In a fast-moving cafeteria, a customer enters their UPI PIN and their bank account is debited ₹120. However, the order is **NOT placed** (e.g. phone lost internet right after UPI authorization, customer closed the app, or the 10-minute checkout timer expired while paying).

1. Paytm Gateway State:
Transaction is marked `TXN_SUCCESS`. Funds reside securely at Paytm.

≠

2. V Foods Order State:
Order status remains `awaiting_payment` or `cancelled`. No food token issued.

Official Refund Options Provided in Paytm Documentation

Option A: Automated Server-to-Server Refund API

Documented by Paytm as the programmatic reversal channel:

- **Endpoint:** `POST https://securegw.paytm.in/refund/apply`
- **How it works:** The merchant server sends a signed JSON request with Merchant ID, Transaction ID, and a unique Refund ID.
- **Paytm Action:** Paytm automatically reverses the funds to the customer's source account (UPI / Bank / Card).
- **V Foods Usage: Primary method used** for automated and one-click refunds.

Option B: Paytm Merchant Dashboard (Manual Portal)

Documented by Paytm as the administrative backup channel:

- **Portal:** `https://dashboard.paytm.com`
- **How it works:** Merchant logs into the Paytm web portal, searches the Order ID / Transaction ID, and clicks "Initiate Refund".
- **Paytm Action:** Paytm processes the refund from the web interface.
- **V Foods Usage:** Super Admin emergency backup. (Store managers do not need this because V Foods automates it via Option A).

THE V FOODS REFUND LIFECYCLE FOR UNPLACED ORDERS

Customer Debited ₹120
Paytm has funds; Order unplaced
→
System Detects Issue
Auto Reconcile / App / Tablet
→
Calls Paytm Refund API
POST /refund/apply with unique refId
→
Customer Receives 100%
Credited back to Bank / UPI

Key Takeaway for Investors & Store Owners: Because everything is in one PostgreSQL database, inventory and payment states are always clear. When money is taken without an order being placed, V Foods uses Paytm's automated Refund API to send the money back immediately.

How V Foods Triggers the Paytm Refund API

The three real-world execution paths that protect customer money



Trigger 1: Automatic Background Reconciler

When: The customer walked away or closed their browser without returning to V Foods.

- A background task (`/admin/payments/reconcile`) scans for stuck payments older than 2 minutes.
- It queries Paytm's Transaction Status API. If Paytm confirms payment was taken but the 10-minute order timer expired, it **automatically initiates the refund**.
- **Human effort required: ZERO.**



Trigger 2: Customer Taps "Check Status" in App

When: The student is looking at their phone and sees their screen stuck on "Payment Pending".

- Customer taps **"Check Payment Status"**.
- Backend queries Paytm live. If payment succeeded but order window expired, the backend **initiates the refund on the spot**.
- Student's screen instantly displays: "*100% Refund Initiated (Ref: refund-482)*".



Trigger 3: Store Manager One-Click Refund

When: Student shows their bank SMS at the counter saying money was debited without a token.

- Manager opens the **"Payments Needing Attention"** screen (`/admin/payments/attention`).
- Manager sees the transaction with order unplaced.
- Manager taps **"Cancel & Refund"**. Backend calls Paytm Refund API immediately.

Where Does the Money Go & Settlement Speed?

Customer Payment Source	Where the Money Returns	Turnaround Time
UPI (GPay / PhonePe / Paytm)	Directly credited back to the customer's **original linked bank account** via NPCI.	INSTANT TO A FEW HOURS
V Foods In-App Wallet	Directly re-credited to the customer's **in-app wallet balance**.	INSTANT (0 SECONDS)
Debit / Credit Card	Credited back to the card account through the card network (Visa / Mastercard / RuPay).	1 TO 3 BANKING DAYS
Netbanking	Credited back to customer's savings account via bank IMPS/NEFT reversal.	2 TO 4 BANKING HOURS

Behind the Scenes: The V Foods Python Backend Function

In `backend/main.py`, line 459, the backend executes:

```
async def _paytm_initiate_refund(payment_id, amount, reason, order_id):
```

- Generates a deterministic reference: `refId = f"refund-{payment_id}"`.
- Calculates HMAC-SHA256 signature using the secret merchant key.
- Transmits HTTP POST to `https://securegw.paytm.in/refund/apply`.
- Updates the PostgreSQL `refunds` table to record the gateway refund ID.

Official Paytm Documentation Proof & API Fields

Visual proof from Paytm's official Developer Documentation portal

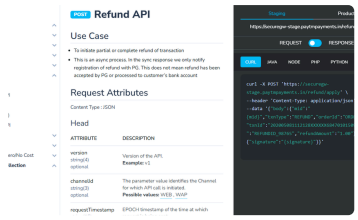


Figure 1: Paytm Official Refund API Specification (POST /refund/apply)
Official Documentation URL: https://paytmpayments.com/docs/api/refund-api
Shows the exact merchant initiation endpoint, required parameters, and response structure.

Exact JSON Sent to Paytm's Refund API

```
{
  "head": {
    "clientId": "C11",
    "signature": "HMAC_SHA256_CHECKSUM..."
  },
  "body": {
    "mid": "YOUR_MERCHANT_ID",
    "txnType": "REFUND",
    "orderId": "order_abc123_169...",
    "txnId": "20261011111212800110168...",
    "refId": "refund-482",
    "refundAmount": "120.00"
  }
}
```

Purpose of Parameters:

- `txnId` : Points to the exact original payment.
- `refId` : Unique refund identifier generated by V Foods.
- `refundAmount` : Exact amount to return.

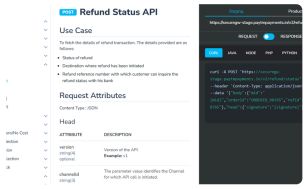


Figure 2: Paytm Refund Status API (/v2/refund/status)
Official Documentation: paytmpayments.com/docs/api/refund-status-api
Used to query refund progress if gateway experiences banking delays.



Figure 3: Paytm S2S Refund Webhook
Official Documentation: paytmpayments.com/docs/accept-refund-webhook
Paytm pushes async status when funds clear into the customer's bank.

Safety Guarantees & Executive Summary

How duplicate refunds are prevented and why this architecture guarantees zero financial loss

What Prevents Duplicate / Double Refunds?

A common concern is: "*What if the student taps 'Check Status' 5 times, or both the automated system and the store manager click Refund?*"

1. Deterministic Key:
V Foods generates: `refId = refund-<payment_id>`. There is only one valid refund key per payment ID.

2. Database Constraint:
In PostgreSQL, `refunds.refund_ref` is marked `UNIQUE`. Inserting a duplicate key raises a constraint error.

3. Paytm Gateway Lock:
Paytm uses `refId` as an **idempotency key**. If Paytm sees the same `refId` again, it returns the existing status without debiting the merchant twice.

Stakeholder Experience Matrix

Stakeholder	What They Experience When an Order Fails	Why It Builds Trust
The Customer (Student)	Screen shows clear status: "*Order unplaced. 100% refund of ₹120 initiated to UPI (Ref: refund-482)*".	No panic, no yelling at cashiers, instant digital proof in the app.
The Store Manager (Cashier)	Admin Tablet highlights unplaced orders under "Payments Needing Attention". One tap cancels & refunds.	No manual cash payouts, no passbook reconciliation headaches, no lost food.
The Investor / Owner	Zero financial leakage. All reversals happen through regulated Paytm banking APIs with auditable logs.	Scalable campus operations without human error or employee fraud.

Final Conclusion:
When payment is debited but an order is not placed, V Foods uses **Paytm's official automated Refund API (`/refund/apply`)**. Whether triggered by the **automatic background worker**, the **customer's in-app status check**, or the **store manager's counter tablet**, 100% of the money is returned safely to the customer's original source account with zero manual bank transfers required.